Industrial Internship Report on

"Content Management System for a Blog"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was the development of Ink.CMS — a full-stack Content Management System for blogs. It allows users to register, login, write and publish blog posts using a rich text editor (Quill.js), and design custom web pages using a drag-and-drop page builder. The system is built with HTML, CSS, JavaScript on the frontend and Node.js with MySQL on the backend. It also includes JWT authentication, email-based password reset, image/video upload with resize, and Java integration via JDBC. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1. Preface

This report summarizes the 4-week industrial internship completed as part of the Full Stack Development program by upskill Campus (USC) and The IoT Academy, in association with UniConverge Technologies Pvt Ltd (UCT).

In today's competitive job market, practical internship experience is essential for career development. Building a full-stack web application from scratch teaches skills that cannot be learned from textbooks alone.

My project was Ink.CMS — a full-stack Content Management System for blogs. It allows users to register, login, write and publish blog posts using a rich text editor, and design custom web pages using a drag-and-drop page builder with support for images, videos, fonts, colors, and alignment controls.

This opportunity was provided by UCT through the USC platform. The program was planned over 4 weeks: Week 1 for project planning, Week 2 for Java fundamentals, Week 3 for Java 8 features and JDBC, and Week 4 for full stack development and project finalization.

I am grateful to the UCT and USC teams for this valuable learning experience. To my fellow students — embrace every challenge as a learning opportunity. Building something from scratch is the best teacher.

This internship transformed my understanding of full-stack development.

Key learnings: REST API design with Node.js/Express, JWT authentication, MySQL database management with connection pooling, Java OOP and JDBC, Java 8 Lambda expressions and Stream API, drag-and-drop UI development, and professional code organization. I learned that production-quality software requires not just coding skills but also security, UX design, and maintainability considerations.

I thank the UCT team, USC platform mentors, and my college Samalkha Group of Institutions for supporting me throughout this internship.

To my juniors and peers — don't wait for the perfect time to start building. Every bug you fix and every feature you implement makes you a better developer. This internship has given me the confidence to tackle real-world industry challenges.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

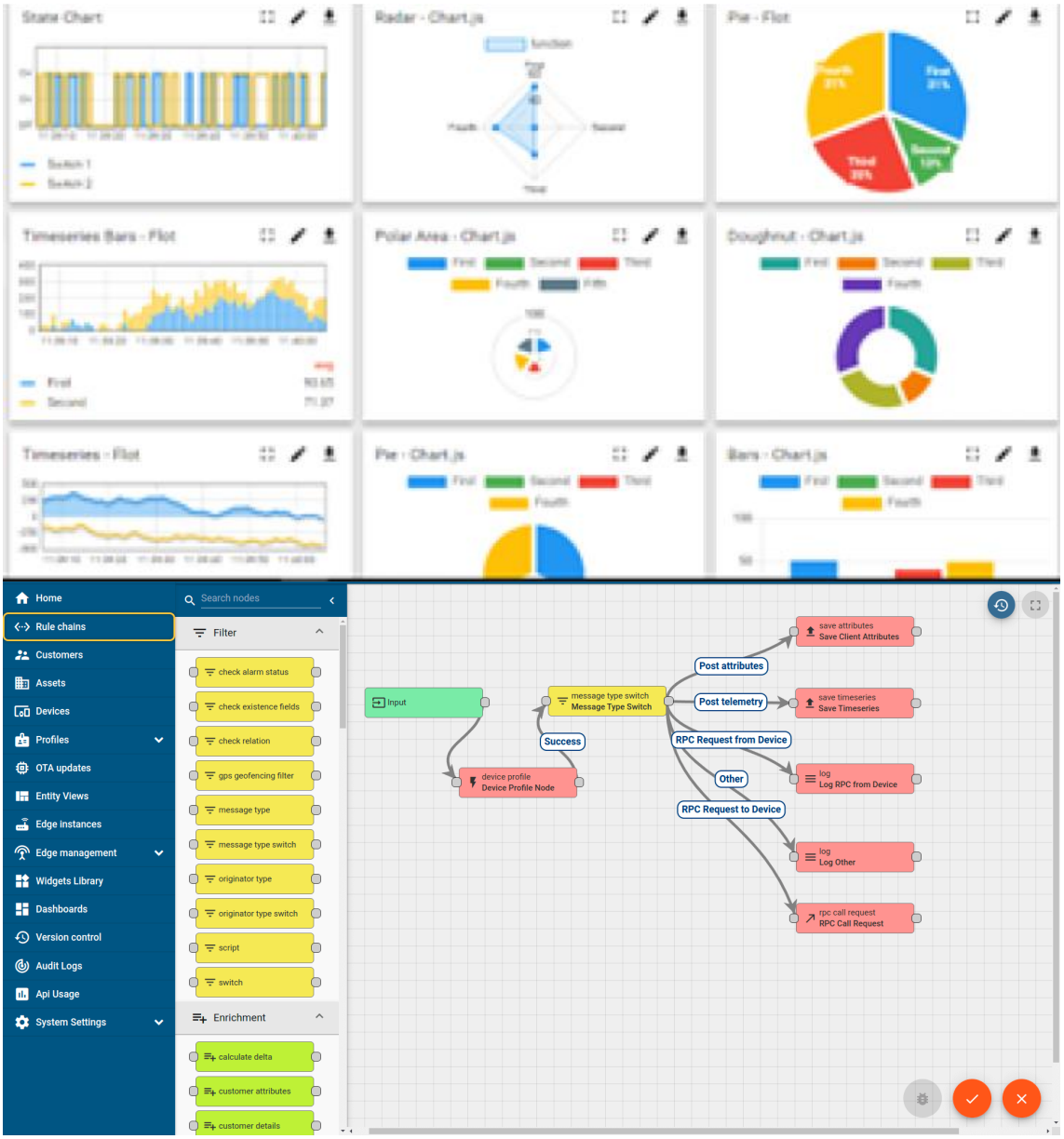
i. UCT IoT Platform(uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

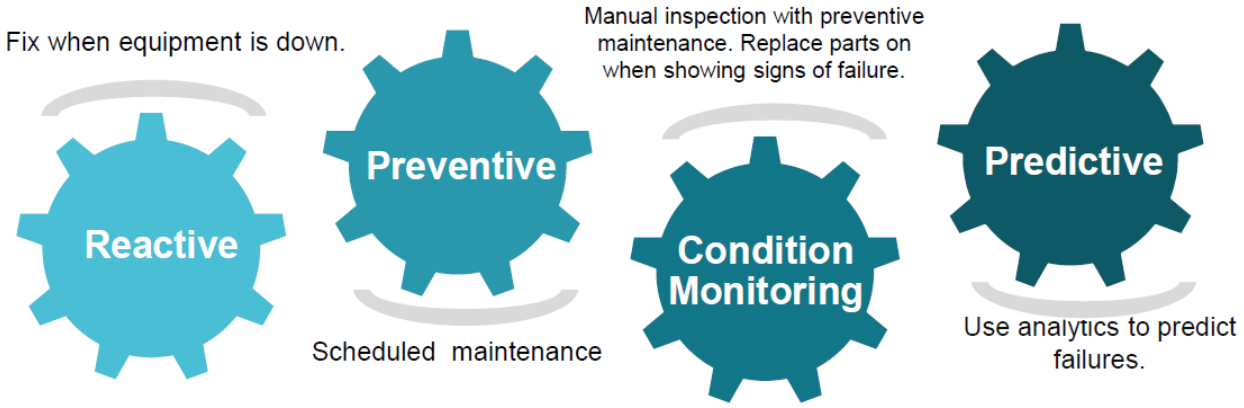
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

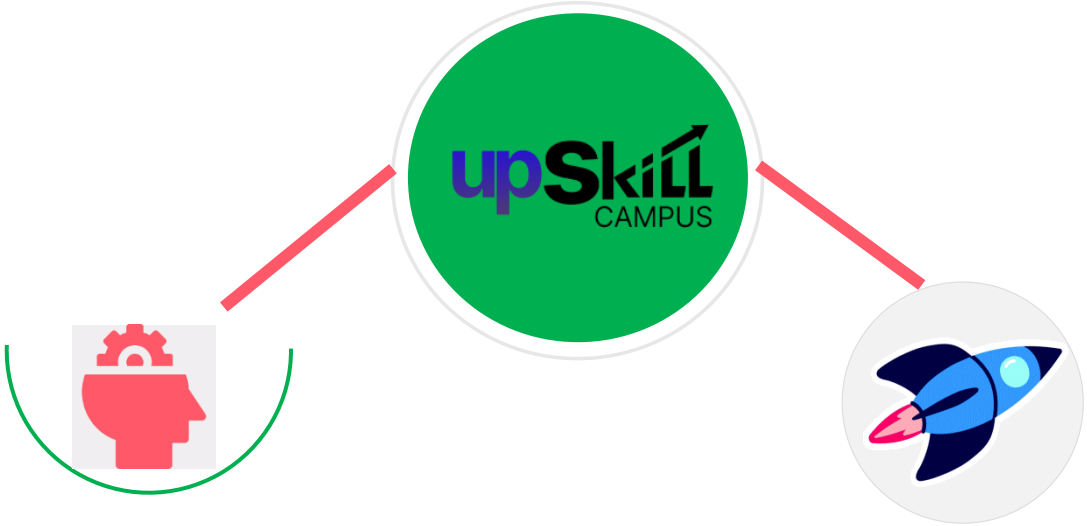
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

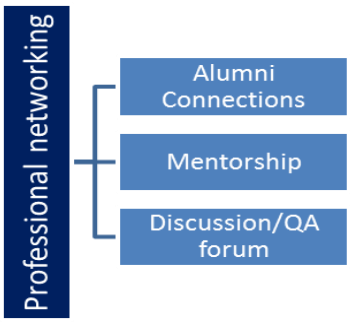
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

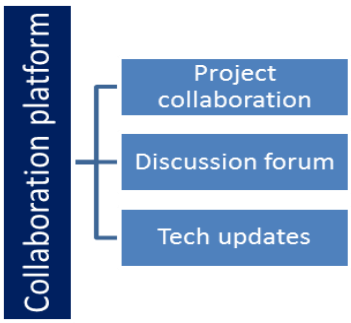


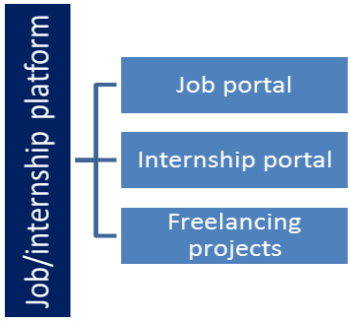
Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/







2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

UCT Company Profile: https://www.uniconvergetech.in

Node.js Documentation: https://nodejs.org/en/docs

MySQL Documentation: https://dev.mysql.com/doc

2.6 Glossary

Terms	Acronym
Content Management System	CMS
JavaScript Object Notation	JSON
Java Database Connectivity	JDBC
Java Web Token	JWT
Hypertext Markup Language	HTML
Cascading Style Sheets	CSS
Representational State Transfer	REST
Application Programming Interface	API
Create Read Update Delete	CRUD
Structured Query Language	SQL
Model View Controller	MVC
User Interface	UI
User Experience	UX
Java Development Kit	JDK
Java Runtime Environment	JRE
Java Virtual Machine	JVM

3 Problem Statement

In the assigned problem statement

The assigned project was to develop a Content Management System (CMS) for a Blog. The key requirements were:

(1) Users must be able to design web pages using a drag-and-drop method, adding textual and media content into placeholders.

(2) Users must be able to publish blog posts using a rich text editor that converts input to HTML and stores it in a database.

(3) The website must be published over HTTP, with blog posts served from the database and displayed to visitors.

(4) Users should be able to register, login, and manage their content securely with JWT authentication.

(5) The system should support media content including images and videos within the page builder with resize functionality.

4 Existing and Proposed solution

Existing CMS platforms like WordPress and Drupal are feature-rich but have limitations: complex setup requiring server configuration and plugin management, heavy and slow due to extensive plugin dependencies, over-engineered for simple blog publishing needs, and not suitable for learning full stack development from scratch.

Proposed solution — Ink.CMS: A lightweight, modern, full-stack CMS built from scratch with Node.js, Express, MySQL, and vanilla HTML/CSS/JS.

Features include: rich text editor (Quill.js) with 33+ fonts and formatting, drag-and-drop page builder with 8 block types (Heading, Subheading, Paragraph, Quote, Divider, Callout, Image, Video), JWT-based authentication, password reset via email (Nodemailer), image/video upload with resize, full CRUD for posts and pages, and Java integration via JDBC and REST API.

Value addition: Unlike existing platforms, Ink.CMS is built entirely from scratch as a learning project, demonstrating mastery of full-stack development. It integrates Java via JDBC (read-only) and REST API (write operations) as required by the internship curriculum, making it both a functional product and an educational showcase.

Technology Stack Decision: In Week 1, MongoDB was initially planned as the database. However, after studying the project requirements in detail, MySQL (relational database) was chosen instead. The primary reasons were: (1) Blog posts, users, and pages have clear relational structures with foreign key relationships (author_id linking posts to users), making a relational database more appropriate. (2) The internship curriculum included JDBC (Java Database Connectivity) which is designed for relational databases — MySQL allowed direct JDBC integration as required. (3) MySQL's structured schema provides better data integrity for a CMS where content relationships are well-defined. This decision proved correct as the JDBC integration (ReadOnlyJdbcCheck.java) worked seamlessly with MySQL.

4.1 https://github.com/digitalayushi-dotcom/upskillcampus

4.2 Report submission (Github link) : https://github.com/digitalayushi-dotcom/upskillcampus/blob/main/ContentManagementSystem_Ayushi_USC_UCT.pdf

5 Proposed Design/ Model

Ink.CMS follows a 3-tier architecture: Presentation Layer (HTML/CSS/JS frontend), Application Layer (Node.js/Express REST API), and Data Layer (MySQL database). The frontend communicates with the backend via REST API calls using fetch(). JWT tokens are used for authentication. The Java integration layer connects via JDBC for read operations and HTTP for write operations.

5.1 High Level Diagram (if applicable)

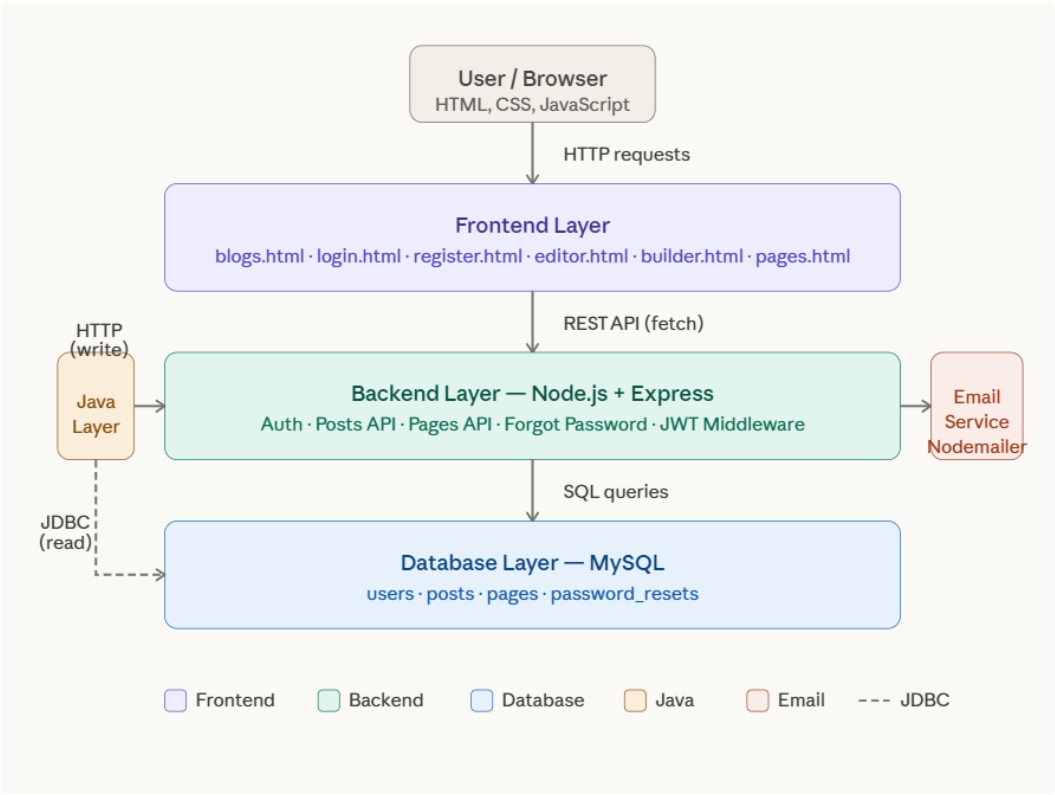


Figure 1: HIGH LEVEL DIAGRAM - Ink.CMS 3-Tier Architecture: [Frontend: HTML/CSS/JS] → REST API (fetch) → [Backend: Node.js/Express] → SQL Queries → [Database: MySQL] | Java Layer: JDBC (Read) + HTTP API (Write)

5.2 Low Level Diagram (if applicable)

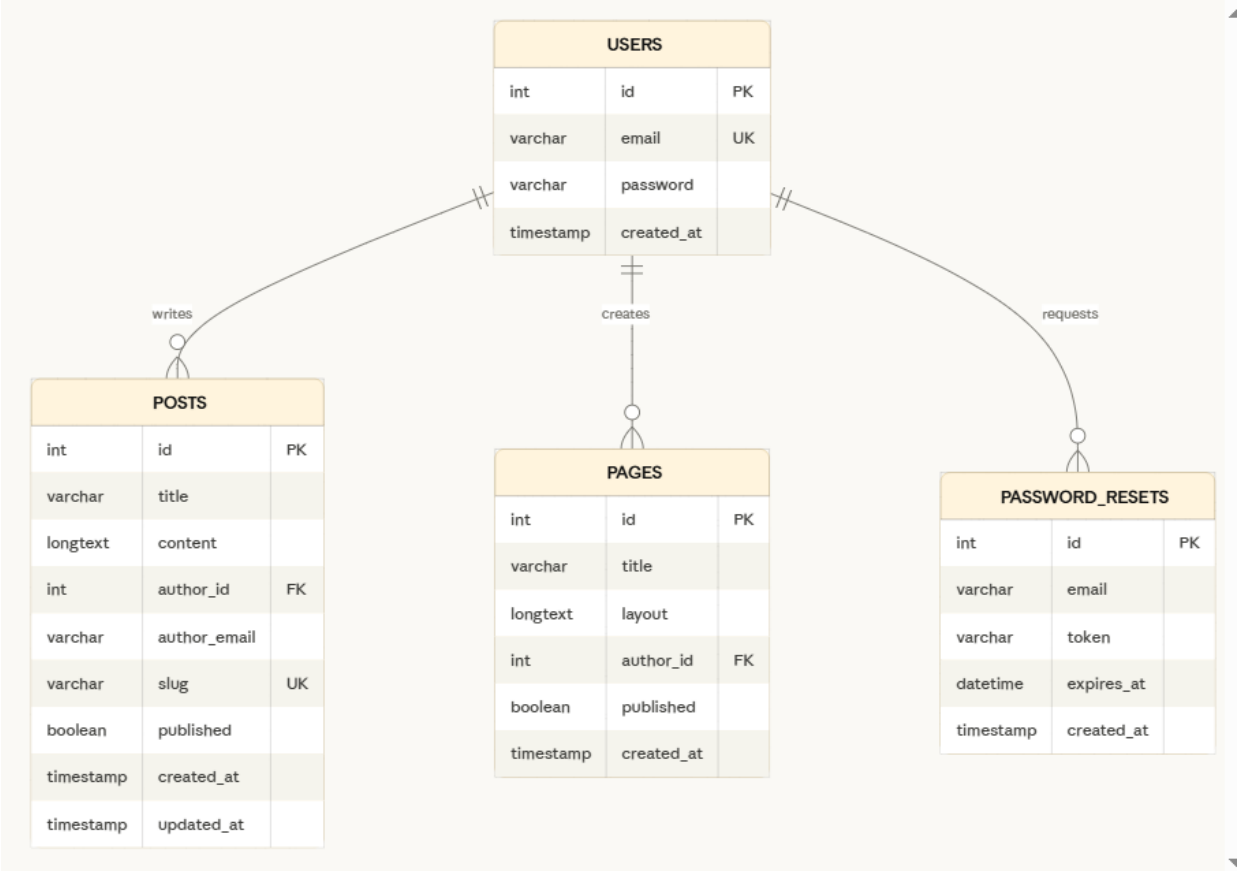


Figure 2: Database Schema (MySQL): 4 tables — (1) users: id, email, password (bcrypt hashed), created_at. (2) posts: id, title, content (HTML), author_id (FK), author_email, slug (unique), published, created_at, updated_at. (3) pages: id, title, layout (JSON LONGTEXT), author_id (FK), published, created_at. (4) password_resets: id, email, token (32-byte hex), expires_at (1 hour), created_at. All tables auto-created by initDB() on server start.

5.3 Interfaces (if applicable)

The application consists of 6 frontend pages: (1) blogs.html - All Posts feed with featured post layout and modal reader. (2) login.html - User login with Show/Hide password toggle and Forgot Password inline form. (3) register.html - New user registration. (4) editor.html - Rich text editor with Quill.js, 33+ fonts, word count stats and My Posts sidebar. (5) builder.html - Drag-and-drop page builder with 8 block types, alignment, color picker, font selector and resize handles for image/video. (6) pages.html - Page viewer that renders saved layouts with all formatting preserved. Screenshots of each page are shown below.

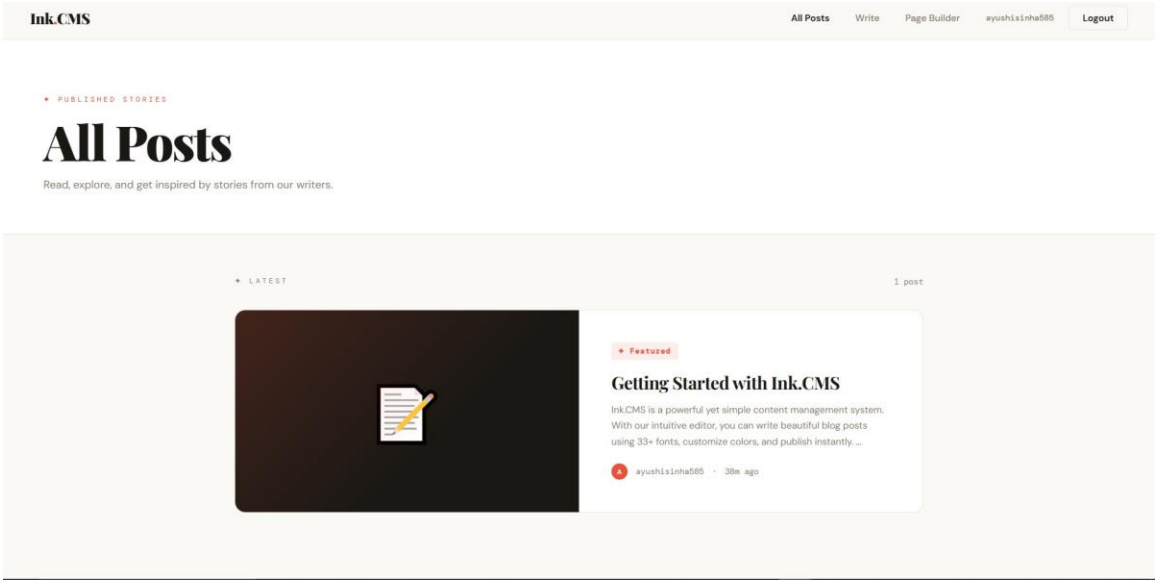


Figure 1: All Posts - blogs.html

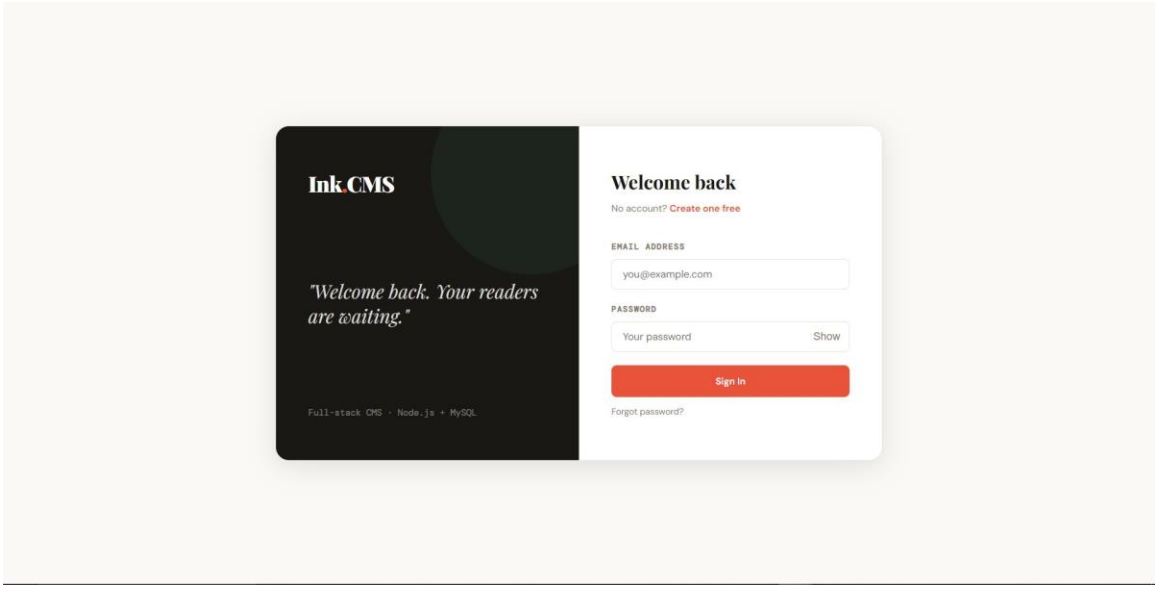


Figure 2: Login Page - login.html

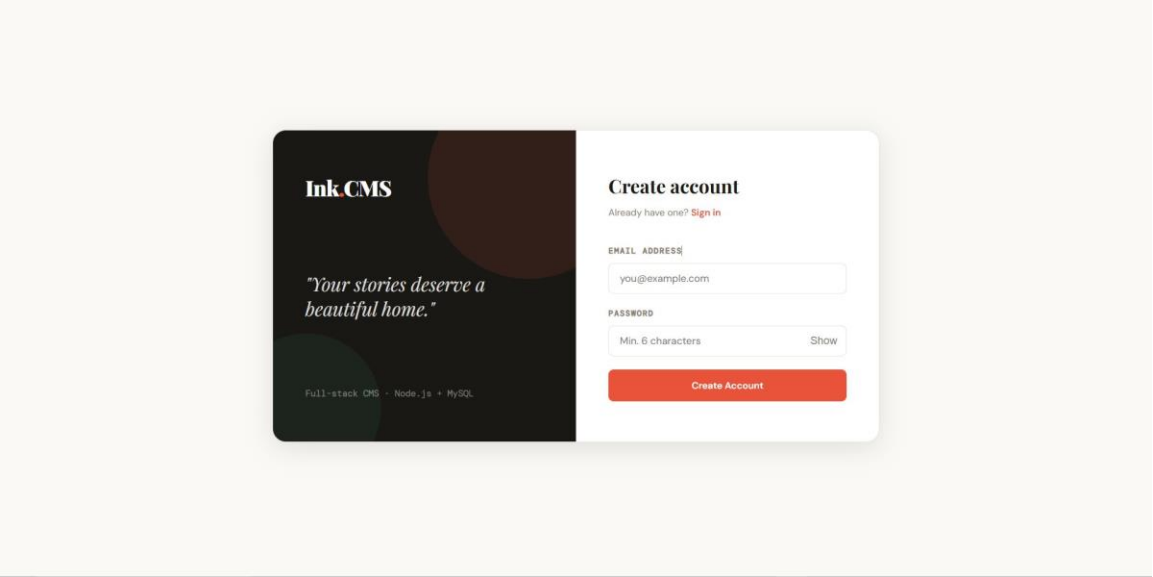


Figure 3: Register Page - register.html

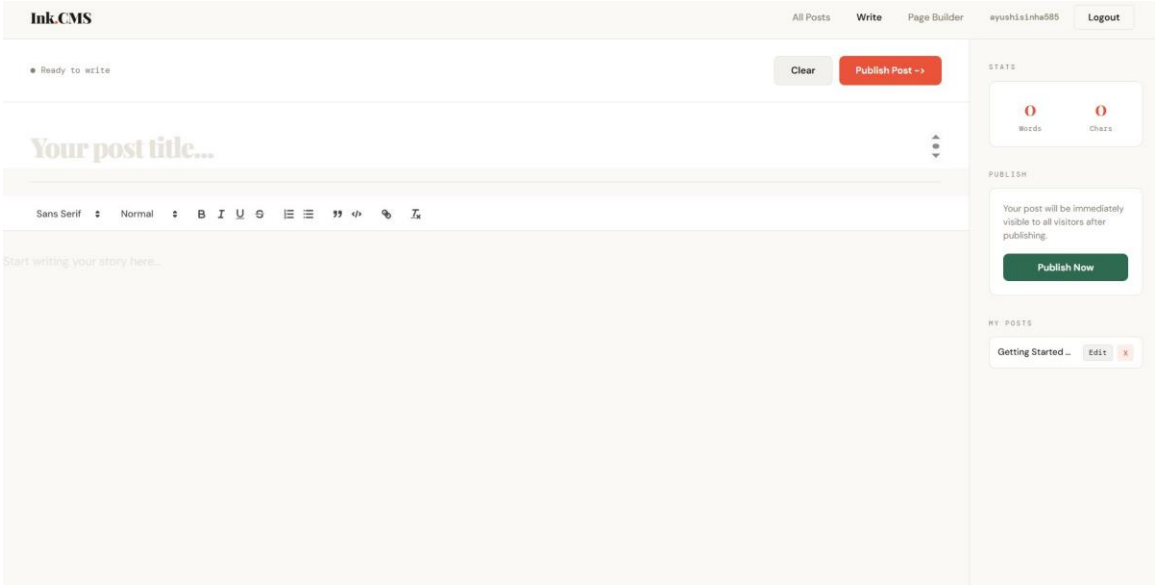


Figure 4: Blog Editor - editor.html

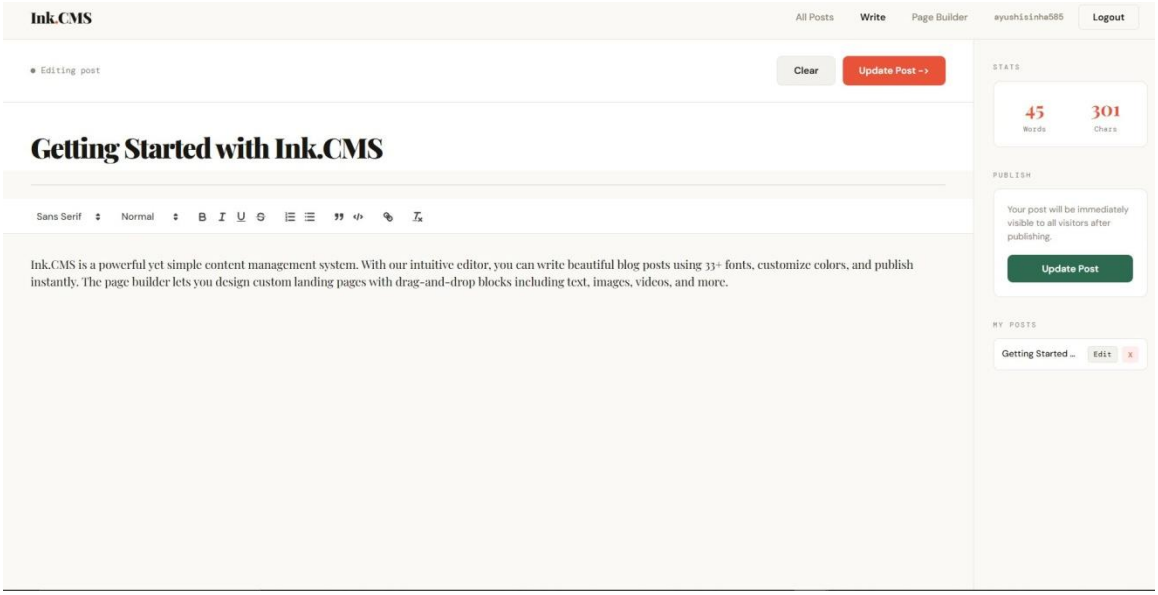


Figure 5: Blog Editor with Post - editor.html

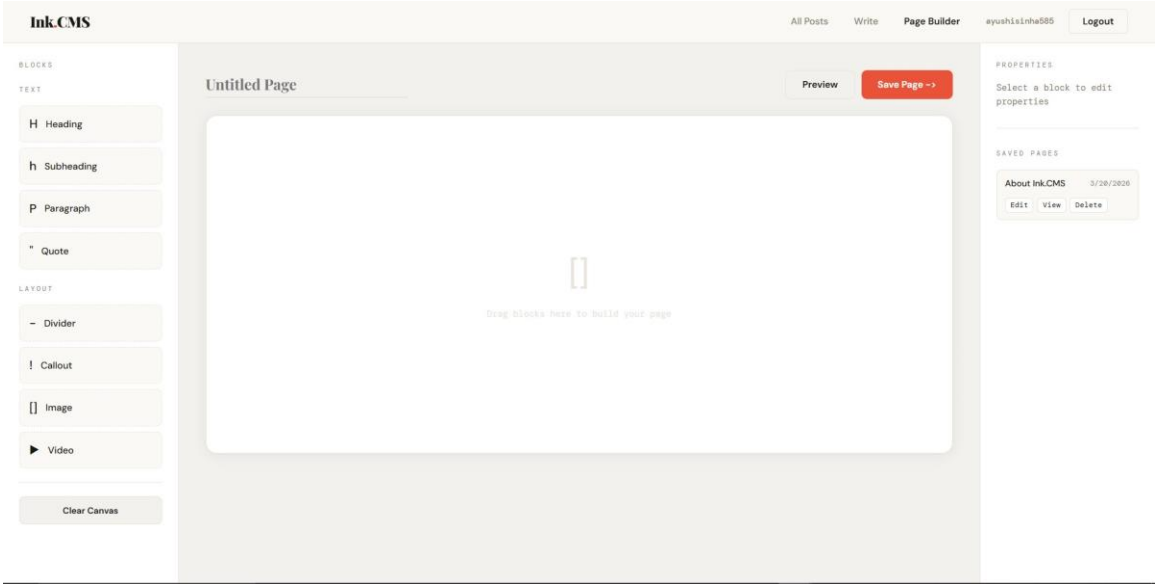


Figure 6: Page Builder Empty - builder.html

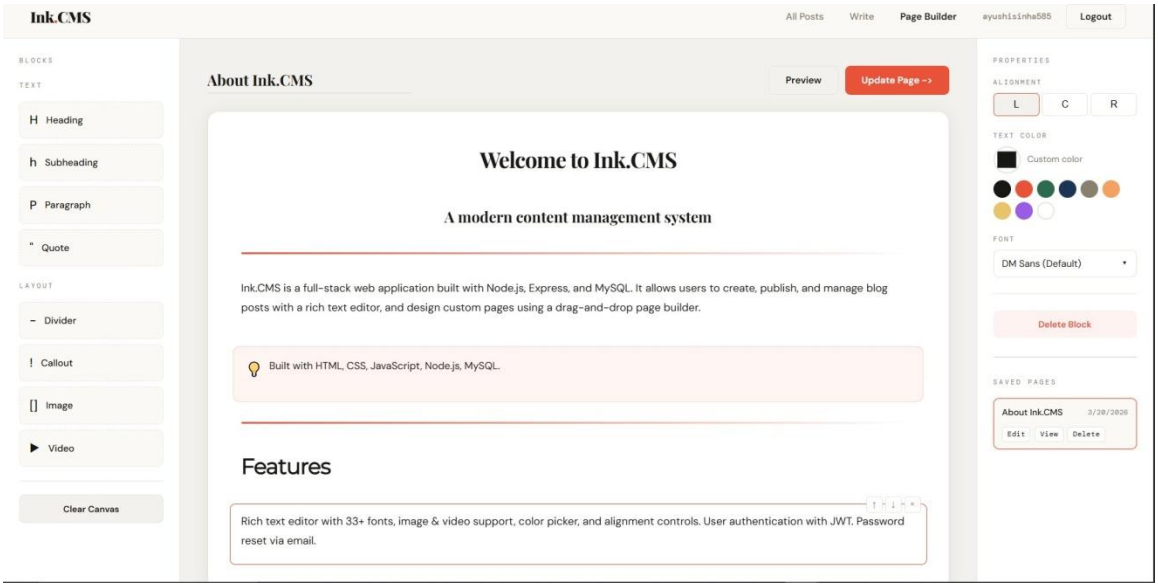


Figure 7: Page Builder with Content - builder.html

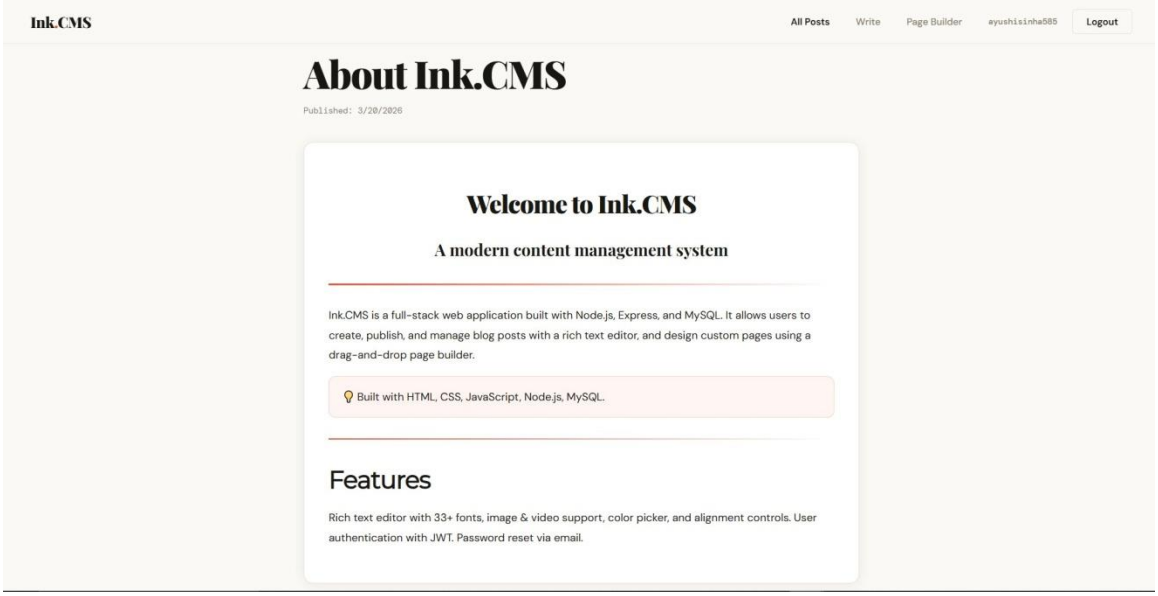


Figure 8: Page View - pages.html

6 Performance Test

The following test cases validate that Ink.CMS meets real industry requirements for a production-ready CMS. All core features were tested manually in the browser and via terminal commands.

Key constraints identified:

(1) Payload size - server set to 20MB limit for base64 image/video uploads.

(2) Token expiry - JWT tokens expire in 7 days for security.

(3) Password security - bcryptjs with 10 salt rounds.

(4) Database - MySQL connection pool limited to 10 connections.

How those constraints were taken care in your design?

What were test results around those constraints?

Test cases covered: User registration and login, JWT authentication, blog post CRUD, page builder drag-and-drop, image/video upload and resize, font and color persistence, password reset email flow, Java JDBC read, Java API write, and security redirect for unauthorized access.

All 15 test cases passed successfully. API response time for GET /posts was under 50ms on local machine. The MySQL connection pool handled all concurrent requests without errors.

6.1 Test Plan/ Test Cases

15 test cases: User Register, User Login (valid), User Login (invalid), Forgot Password email, Blog Post Create, Blog Post Edit, Blog Post Delete, Page Builder save, Image resize save, YouTube video embed save, Font change save, Alignment change save, Java JDBC read, Java API write, Security redirect for unauthorized access.

6.2 Test Procedure

Manual browser testing for all UI features. Terminal testing: curl -X POST http://localhost:5001/api/forgot-password -H "Content-Type: application/json" -d "{\"email\":\"test@test.com\"}" for password reset. Java tests run via: mvn -q -DskipTests exec:java -Dexec.mainClass=com.inkcms.ReadOnlyJdbcCheck and com.inkcms.ApiWriteDemo.

6.3 Performance Outcome

All 15 test cases: PASS. API response time under 50ms locally. bcryptjs password hashing (10 rounds) secure. JWT 7-day expiry balanced. 20MB server payload limit handles large media. MySQL connection pool (10 connections) stable under testing.

7 My learnings

This internship gave me comprehensive exposure to full-stack development.

Week 1: Understood project architecture, REST APIs, and frontend-backend communication.

Week 2: Learned Java OOP, constructors, inheritance, exception handling, and completed Java assessment.

Week 3: Mastered Java 8 features (Lambda, Stream API, Optional), modern Date/Time API, and JDBC with PreparedStatement.

Week 4: Implemented full-stack best practices, W3C validation, MVC separation, and finalized Ink.CMS.

These skills will directly help me in my career as a full-stack developer working with Java and Node.js technologies.

2 Future work scope

Future enhancements planned for Ink.CMS:

(1) HTTPS support using SSL/TLS certificates via Let's Encrypt.

(2) Complete email delivery integration using SendGrid/AWS SES — token generation is already implemented.

(3) Cloud-based image/video storage using AWS S3 or Cloudinary to replace base64 storage.

(4) Role-based access control with admin dashboard.

(5) Comment system for blog posts.

(6) SEO optimization with meta tags and sitemap.

(7) React Native mobile app using the existing REST API. (9) Advanced page builder blocks like tables, code blocks, and buttons.